

- Use the object **Exporter**  for providing and exporting services. It represents a group of people, whose individual members you cannot distinguish and whom you cannot address as individuals.
- Use the object **FootPath**  for modeling a path on which the *Worker* walks from the *WorkerPool* to the *Workplace*.
- Use the object **LockoutZone**  for controlling a group of material flow objects. If one of these objects fails, all other objects within the lockout zone stop processing parts
- Use the object **Marker**  for setting milestones along which the AGVs drive from the *AGVPool* to their destination.
- Use the object **ShiftCalendar**  for modeling a shift work system in your plant.
- Use the object **Worker**  for doing a job on a workplace at a station.

The toolbar **Resources** does not show the *Worker*. You can access the class of the *Worker* in the *Class Library*.

- Use the object **WorkerPool**  for modeling the staff room of your plant.
- Use the object **Workplace**  for modeling the actual place attached to the station, where the *Worker* performs his job.

Do not confuse the *Plant Simulation* **resource objects** with the *Plant Simulation* **material flow resources**. The **material flow resources** are any of the built-in **Material Flow Objects**, **Fluid Objects** or a *Frame* in which you modeled a machine.

Workplace

Workplace

Use the object *Workplace* for modeling the actual place at the station, where the *Worker* performs his job.

Image



Description

The *Worker-WorkerPool-Workplace-FootPath*-concept refines the *Broker-Importer-Exporter*-concept. You can assign a *Workplace*  to the material flow objects that support an *Importer*. In this case, the importers are **processing** or **repairing importers**.

For the **transport importer**, who controls how the *Workers* carry parts, you can assign a *Workplace* to the **Buffer**, the **PlaceBuffer**, the **Source**, the **Sorter**, and the **Store** in addition.

The *Workplace* is the actual place at the station where the *Worker* performs his job.

Note

You can, but you do not have to assign a *Workplace* to a station, i.e., a material flow object.

If you do not use a **FootPath**, *Plant Simulation* transports the *Worker* directly to the *Workplace*. Here the *Worker* functions as an **Exporter** with a capacity of 1.

The *Workers* are created in the **WorkerPool** and they stay there when they do not work and are waiting for a job order.

The **Broker** procures the *Worker* for the individual work stations. As soon as a *Worker* can provide a service, the *Broker* sends him from his *WorkerPool* to the work station that ordered him:

- If the *Worker* walks freely within the area of the simulation model, he walks on the shortest route to the *Workplace* and walks around any obstacles you designated.

The *Worker* consumes time, when he walks freely within the area to the *Workplace*.

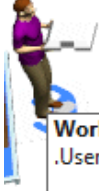
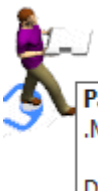
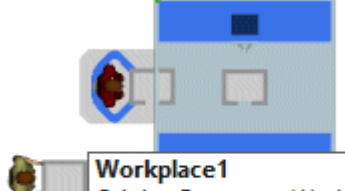
- If you modeled the *work station* using one of the material flow objects and the *Workplace* and connected the *WorkerPool* and the *Workplace* with a *FootPath*, *Plant Simulation* animates the *Worker* while he moves on the *FootPath* and while he works on the *Workplace*.

The *Worker* consumes time when he walks on the *FootPath*.

If several *Workers* walk to the same *Workplace* and have to wait their turn before they can step onto it, you can define **how they queue up in front of the workplace**.

- If you activated **Beam to workplace** and if you inserted a *Workplace*, and the *Worker*, who is supposed to work there, cannot reach it via a *FootPath*, *Plant Simulation* beams the *Worker* directly to the *Workplace* and animates him there.
- If you activated **Workers can work remotely**, if you did not insert a *Workplace*, if no *Workplace* exists that supports this service, or if all *Workplaces* that support this service are occupied, *Plant Simulation* seemingly transports the *Worker* to the work station itself. He then does his job at the work station. *Plant Simulation* does not animate him at the work station though, but continues to animate him in the *WorkerPool*.

To show a **tooltip** with information about the *Workplace*, hover with the mouse over it.

Part Carried by Worker with Route	Part Carried by Worker Destination	Workplace Assigned to Station Workplace Reserved for Worker (filled Arrow) Workplace Reserved for Worker on his Way (hollow Arrow)
 Worker2 UserObjects.Worker2:1 Route (3.7m): root.Workplace root.Workplace1 Next event: PositionReached (+2.4505)	 Part .MUs.Part:2 on .UserObjects.Worker2:1 Destination = root.Station	 Workplace1 Origin: .Resources.Workplace ⇒ Station @ Entrance ★ .UserObjects.Worker3:1

To change the length of the graphic and the anchor points of the *Workplace*, click [Show Manipulators](#)



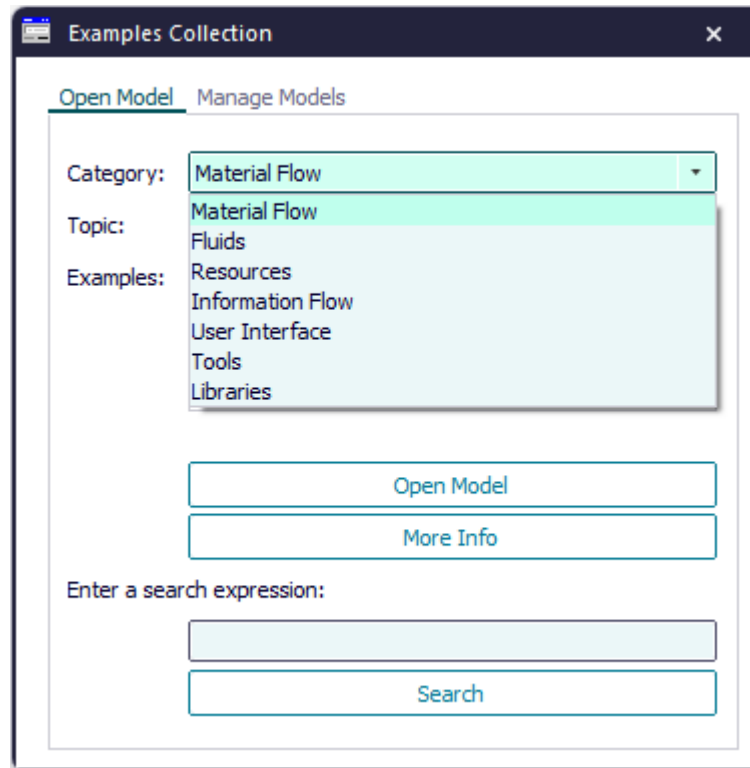
on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model



To add the object *Workplace* to your simulation model, click [Manage Class Library](#) > **Basic Objects** > **Resources** > **Workplace** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model Workers and the Jobs They Do](#)

[How Workers Queue Up in Front of the Workplace](#)

[Model a Worker Who Waits in Line](#)

[Tab Importer \[general description\]](#)

[Worker-WorkerPool-Workplace-FootPath](#)-concept

[Broker-Importer-Exporter](#)-concept

[Video on YouTube](#)

https://youtu.be/HiPziA8kxc0?si=Ft4r3_Sq9xlzt9DU&t=45

How Workers Queue Up in Front of the Workplace

You can set how arriving *Workers*, who cannot step onto the *Workplace*, wait in the queue in front of it.

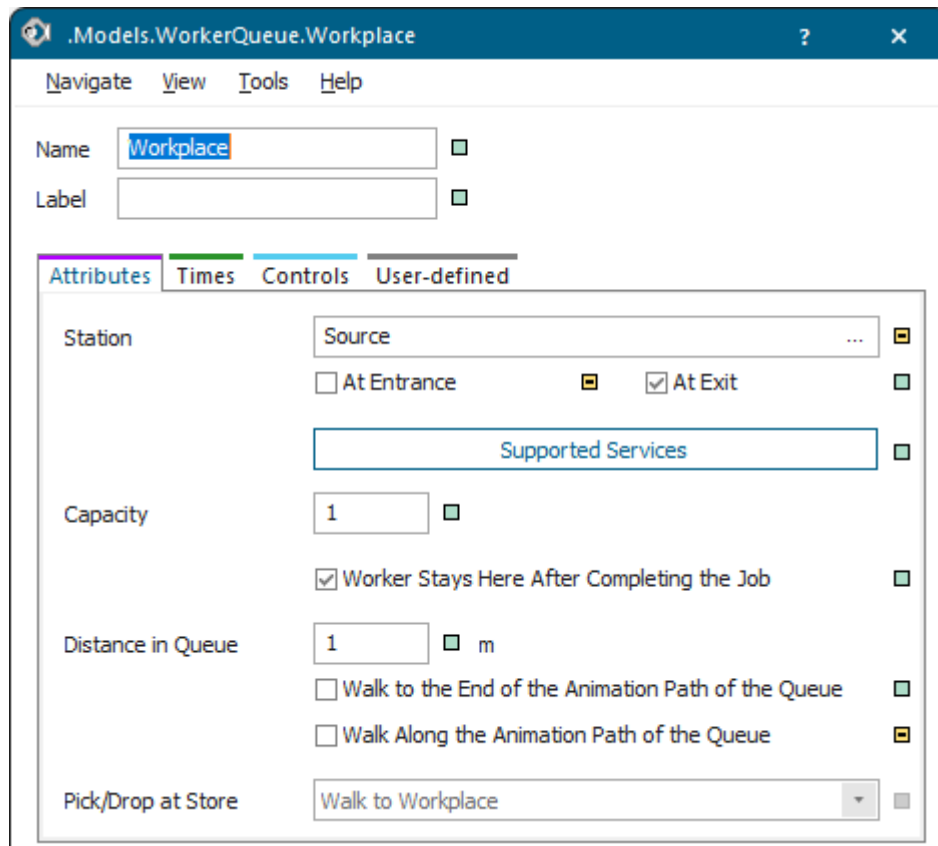
Remarks

Plant Simulation queues these *Workers*, without using additional *FootPaths*, in the queue at their **waiting position**. The **waiting position** is determined according to the position in the **Forward Blocking List** of the *Workplace*. Let's say the *Worker* is located at the third position in the **Forward Blocking List**, then he also waits at the third place in the queue.

The **waiting position** is the position at which the *Worker* waits until it is his turn to step onto the *Workplace*. The **waiting position** depends on his position in the **Forward Blocking List** and the distance between the *Workers* in the queue. Once a waiting *Worker* moves up in the **Forward Blocking List**, he also moves up in the queue.

In new models *Plant Simulation* visualizes the queue on the **MU Animation Path** named **Queue** of the *Workplace*. If the queue is longer than the pre-defined **MU Animation Path**, *Plant Simulation* elongates it toward the end of the queue.

You can set the **Distance** between the *Workers* in the queue on the animation path.



If you do not want the *Worker* to walk to the *Workplace*, but to the end of the **MU Animation Path**, select **Walk Along the Animation Path of the Queue**.

After the *Worker* has reached the end of the **MU Animation Path**, *Plant Simulation* computes the direct way to the **waiting position** or to the *Workplace* respectively.

Working with the Worker Queue

- If the *Worker* is to **Model Workers Who Walk to the End of the Queue**, the end of the animation path has to be positioned at the same height (**Z-position**) as the *Workplace*.

If this is not the case, *Plant Simulation* might not be able to compute a route to the end of the animation path.

Once the *Worker* arrived at the end of the animation path and provided the *Workplace* is still available while computing the route, *Plant Simulation* sets the *Workplace* as the target, even if other *Workers* might already be on their way to the *Workplace* on the animation path.

If the *Worker* arrives at a *Workplace* or at a **waiting position** that is occupied, *Plant Simulation* moves him directly to the correct **waiting position**.

If the end of the animation path is located after the actual **waiting position**, the *Worker* walks along the animation path to this **waiting position**.

If the queue is longer, so that the *Worker* stands in the queue once he has reached the end of the animation path, he walks through the already waiting *Workers* to his **waiting position**.

- Activating **Walk Along the Animation Path of the Queue** and reaching the **waiting position** or the *Workplace* via a *Footpath* is relevant for the simulation as the times during which the *Worker* is on his way do change.

Plant Simulation does not use up any time for reaching the **waiting position** if the *Worker* walks directly to the *Workplace*. *Plant Simulation* then shows him at the respective **waiting position** in the queue.

The same is true when the *Worker* moves up within the queue as these movements are executed by the respective animation of the *Worker*.

See also

Model a Worker Who Waits in Line

Distance in Queue

Walk to the End of the Animation Path of the Queue

Walk Along the Animation Path of the Queue

MU Animation on Animation Paths on the Workplace

MU Animation on Point-Oriented Objects Holding Several Parts

Video on YouTube

<https://youtu.be/gU1pqu7sWA8?si=HgdhN-JsmsY7HQSH&t=775>

Dialog Box of the Workplace

Dialog Box of the Workplace

Double-click the icon of the *Workplace* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Station [Workplace]

The **Station** to which the *Workplace* is going to be assigned is a material flow object that supports an *importer*.

Conveyor	Point-oriented object
Station Conveyor ...  ☒ At Entrance ☐ At Exit 	Station Station ...  ☒ At Entrance ☐ At Exit ☒ At Exit 

Remarks

Click the ellipsis button and select the **Station** to which you want to assign the *Workplace*, in the dialog **Select Object**.

The **Station** is a material flow object that supports an *importer*, for example the [Source](#), the [Drain](#), the [Station](#), the [ParallelStation](#), the [AssemblyStation](#), the [DismantleStation](#), or the [Conveyor](#).

To automatically enter the **Station**, drag the *Workplace* close to a side of the material flow object which you want to use as the **Station**.

The image of the *Workplace*, to which you did not assign a **Station**, looks like this . The image of the *Workplace*, to which you did assign a **Station**, looks like this .

Note

F2 opens the dialog of the object, whose name you typed into the text box.

SimTalk

Station [SimTalk]

See also

Select Object [for controls]

At Entrance [check box]

To attach the *Workplace* to the entrance of the *Conveyor*, select this check box.

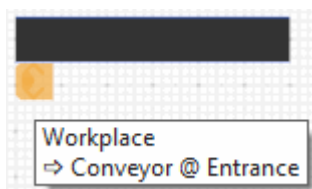
Conveyor	Point-oriented object
Station Conveyor ...  ☒ At Entrance  ☐ At Exit 	Station Station ...  ☒ At Entrance  ☒ At Exit 

Remarks

For the place-oriented material flow objects *Plant Simulation* selects **At Entrance** and **At Exit** by default.

Attach a Workplace to the Entrance of a Conveyor with Drag-and-Drop

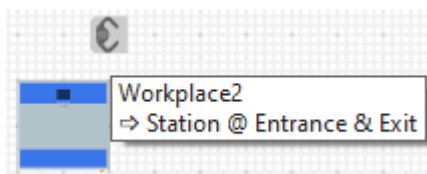
- Select the *Workplace* with the mouse and drag it to the entrance of the *Conveyor*. The icon of the *Workplace* shows a tooltip with **Conveyor @ Entrance**.



- Select the *Conveyor*, hold down **Shift** and drag it to the icon of the *Workplace* to attach it to its entrance.

Attach a Workplace to a Place-oriented Material Flow Object with Drag-and-Drop

- Select the *Workplace* with the mouse and drag it to the object, the *Station* in the example below. The icon of the *Workplace* shows a tooltip with **Station @ Entrance & Exit**.



- Select the *Station* and drag it to the icon of the *Workplace* to attach it to its entrance and exit.

Note

Assigning a new station to a *Workplace* via *SimTalk* does not change the current assignment to the entrance or the exit.

SimTalk

[AtEntrance \[SimTalk\]](#)

See also

[Model a Worker Who Waits for a Free Target](#)

At Exit [check box]

To attach the *Workplace* to the exit of the *Conveyor*, select this check box.

Conveyor	Point-oriented object
Station Conveyor ...  ☒ At Entrance  ☐ At Exit 	Station Station ...  ☒ At Entrance  ☒ At Exit 

Remarks

For the place-oriented material flow objects *Plant Simulation* selects **At Entrance** and **At Exit** by default.

Attach a Workplace to the Exit of a Conveyor with Drag-and-Drop

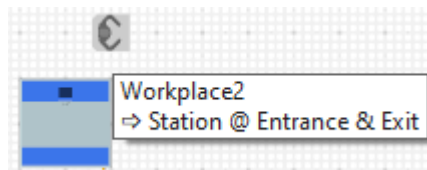
- Select the *Workplace* with the mouse and drag it to the exit of the *Conveyor*. The icon of the *Workplace* shows a tooltip with **Conveyor @ Exit**.



- Select the *Conveyor* and drag it to the icon of the *Workplace* to attach it to its exit.

Attach a Workplace to a Place-oriented Material Flow Object with Drag-and-Drop

- Select the *Workplace* with the mouse and drag it to the object, the *Station* in the example below. The icon of the *Workplace* shows a tooltip with **Station @ Entrance & Exit**.



- Select the *Station* and drag it to the icon of the *Workplace* to attach it to its entrance and exit.

Note

Assigning a new station to a *Workplace* via *SimTalk* does not change the current assignment to the entrance or the exit.

SimTalk

AtExit [SimTalk]

See also

[Model a Worker Who Waits for a Free Target](#)

Capacity [text box] - Workplace

Type the **Capacity** of the *Workplace* into the text box. The **Capacity** is the maximum number of *Workers* that can stay on this *Workplace* at any one time.

Capacity

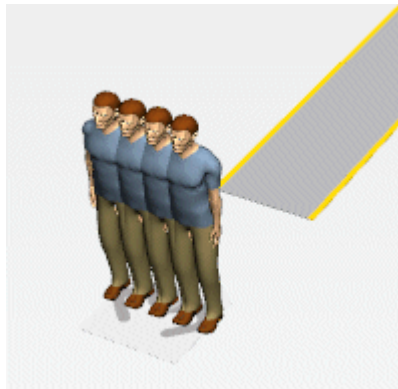
 

Remarks

The **Capacity** is important if several tasks can be accomplished on this *Workplace* or if these tasks come together there.

Plant Simulation uses the coordinate of the *Workplace* to find the shortest route to it, meaning that the *Worker* walks to this coordinate and then jumps to the animation point of the *Workplace*.

Plant Simulation shows the *Workers* on the *Workplace* distributed across an **Animation Area**.



Note

You can only change the **Capacity** if the *Workplace* is empty.

Note

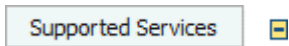
You can define a **Recovery Time** for a *Workplace* that has a **Capacity** of **1**. It starts once a *Worker* leaves the *Workplace*.

SimTalk

Capacity [SimTalk] - Workplace

Supported Services [button]

To open the list into which you type the services, which the *Workplace* supports, click this button.



Remarks

Before you can type in data, click the **Inheritance** check box so that it looks like this .

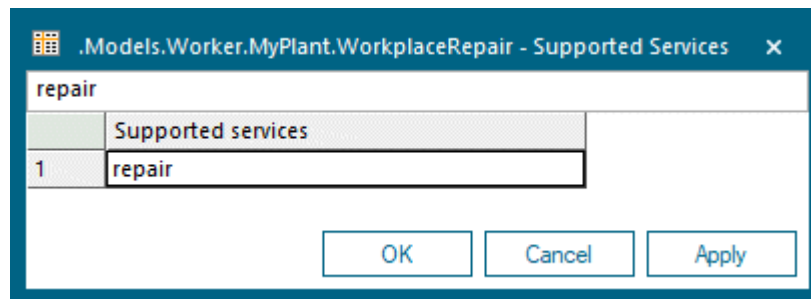
- Type in an expression of your choice into the cell, and click **Apply**. You might, for example, specify different names for the different services, such as **drilling**, **milling**, **turning**, etc. or group them by name.

Note

The name is not case-sensitive, just like the names of *attributes* and *methods* of the objects are not case-sensitive.

To save memory and improve access speed, all places which are using such a case-insensitive string are pointing to the same string in main memory. The visible and unexpected result is that the first occurrence of the string defines how the string is written in terms of upper- and lower-casing.

In *SimTalk* you can compare strings in an case-insensitive manner with the `~=` operator, compare [Relational Operators](#).



- If you do not type any **Services** into the list, the *Worker* can execute any service on this *Workplace*.
- If you did not insert a *Workplace*, if no *Workplace* exists that supports this service, or if all *Workplaces* that support this service are occupied, *Plant Simulation* seemingly transports the *Worker* to the work station itself. He then does his job at the work station.

Note

The *Worker* even works if he does not find a *Workplace* that supports the requested service. In this case, he will not be animated though.

Plant Simulation does not animate the *Worker* at the work station though, but continues to animate him in the *WorkerPool*.

[Supported Services \[button\]](#)

SimTalk

[SupportedServices \[SimTalk\]](#)

See also

[Context Menu of Embedded Lists](#)

Worker Stays Here After Completing the Job

To make the *Worker* stay on the *Workplace* after he completes his job, select this check box.

☒ Worker Stays Here After Completing the Job ☐

Remarks

To make the *Worker* return to the *WorkerPool* or to the **Home Location** immediately after finishing his job, clear the check box.

You have to clear the check box if you activate the setting **Get Job Orders At Home Only** in the *WorkerPool*.

Note

A *Worker*, who is not automatically brokered, stays on the *Workplace* to which he was sent, no matter if you activated **Worker Stays Here After Completing the Job/WorkerStaysHere** or not in the *Workplace*.

Note

If a *Worker* stays on a *Workplace* while another *Worker* has to work on it, and if the **Capacity** of the *Workplace* is used up, *Plant Simulation* shoves the *Worker* aside who is not working. This *Worker* then exits the *Workplace* and returns to the *WorkerPool* or to the **Home Location** if you configured one.

SimTalk

WorkerStaysHere [SimTalk]

See also

Get Job Orders At Home Only

GetJobOrdersAtHomeOnly [SimTalk]

Home Location

Video on YouTube

<https://youtu.be/HiPziA8kxc0?si=fCggWltWi0soTBxq&t=314>

Distance in Queue

Type the **Distance** between the *Workers* in the queue on the animation path of the *Workplace* into the text box. The *Workers* attempt to keep this distance while they wait in line to step onto the *Workplace* and do their job.

Distance in Queue  m

SimTalk

[DistanceInQueue \[SimTalk\]](#)

See also

[How Workers Queue Up in Front of the Workplace](#)

[Model a Worker Who Waits in Line](#)

[Walk to the End of the Animation Path of the Queue](#)

[Walk Along the Animation Path of the Queue](#)

[MU Animation on Animation Paths on the Workplace](#)

Walk to the End of the Animation Path of the Queue

To make the *Worker* first **Walk to the End of the Animation Path of the Queue** and then on to his **destination**, select this check box.

☒ Walk to the End of the Animation Path of the Queue 

SimTalk

[WalkToQueuePathEnd \[SimTalk\]](#)

See also

[How Workers Queue Up in Front of the Workplace](#)

[Model a Worker Who Waits in Line](#)

[Walk Along the Animation Path of the Queue](#)

[MU Animation on Animation Paths on the Workplace](#)

Walk Along the Animation Path of the Queue

To make the *Worker* **Walk Along the Animation Path of the Queue** to his **waiting position** or to the *Workplace* once he has reached the **MU Animation Path**, select this check box.

☒ Walk Along the Animation Path of the Queue 

Remarks

Walk Along the Animation Path of the Queue only takes effect, if you also select **Walk to the End of the Animation Path of the Queue** as the *Worker* first walks to the end of the animation path of the queue and then along the animation path to the *Workplace* or to the **waiting position**.

Once the *Worker* has reached the end of the **MU Animation Path**, *Plant Simulation* computes the route to the target, the **waiting position**, or to the *Workplace* along the **MU Animation Path**.

Note

Plant Simulation does not take obstacles into account when the *Worker* walks along the **MU Animation Path**.

SimTalk

[WalkAlongQueuePath \[SimTalk\]](#)

See also

[How Workers Queue Up in Front of the Workplace](#)

[Model a Worker Who Waits in Line](#)

[MU Animation on Animation Paths on the Workplace](#)

[MU Animation on Animation Paths on the Workplace](#)

Video on YouTube

<https://youtu.be/gU1pqu7sWA8?si=HgdnN-JsmsY7HQSH&t=775>

Pick/Drop at Store

Sets what the *Worker* does when he walks to a *Workplace* attached to a *Store*, when picking up or dropping parts at this *Store*. The *Workplace* only shows this setting if the **Station** is a *Store*.

Remarks

Select how the *Worker* proceeds when he has to pick parts up at the *Store* or when he places parts in the *Store*.

You can select one of these settings:

- **Walk to Workplace**

Is the default setting that previous versions of *Plant Simulation* implicitly provided. The *Worker* walks to the *Workplace* to pick up parts or to drop parts.

For this setting, all settings of the *Workplace* are available.

Attributes	Times	Controls	User-defined
Station	MyStore ...		
	☒ At Entrance	☐	☒ At Exit
	Supported Services		
Capacity	1		
	☒ Worker Stays Here After Completing the Job		
Distance in Queue	1 m		
	☐ Walk to the End of the Animation Path of the Queue		
	☐ Walk Along the Animation Path of the Queue		
Pick/Drop at Store	Walk to Workplace		

Note

Plant Simulation only considers route computations with the methods *getRouteLength* and *getRouteCoordinates* for the setting **Pick/Drop at Store > Walk to Workplace**.

For the other settings of **Pick/Drop at Store** the method *getRouteLength* will return the value -1 and *getRouteCoordinates* will return an empty array.

- **Walk to Store Column**

The *Worker* walks to the store column to pick up or to drop parts. The *Worker* can walk to the store column without any restrictions.

Capacity	-1	☐
	☒ Worker Stays Here After Completing the Job	☐
Distance in Queue	1	☐ m
	☐ Walk to the End of the Animation Path of the Queue	☐
	☐ Walk Along the Animation Path of the Queue	☐
Pick/Drop at Store	Walk to Store Column	

For this setting the following features are not available:

- You cannot enter a **Capacity**.
We assume the **Capacity** to be infinite.
- You cannot select **Worker Stays Here After Completing the Job**.
- You cannot enter the **Distance in Queue**.
- You cannot select **Walk to the End of the Animation Path of the Queue**.
- You cannot select **Walk Along the Animation Path of the Queue**.

Plant Simulation determines the locations to which the *Worker* walks by the dimensions of the side of the *Store* at which the *Workplace* is placed, by the number of columns at that side, and by the distance of the center point of the *Workplace* to the *Store* itself.

In addition, *Plant Simulation* uses the distance of the center point of the *Workplace* to the store front to place the pick or drop locations in front of the store column.

Walk to Store Column uses all pick or drop locations in front of the *Store* for computing the routes of the *Workers*. Be aware that large *Stores* with many columns will increase the complexity of the route computations and will impact performance.

- **Walk Along Store to Store Column**

The *Worker* walks to the first or last column at the side of the *Store* where the *Workplace* is placed. Then the *Worker* extends his route and continues to walk along the *Store* to the store column to drop a part or pick up a part. Thus the routes to the store columns are restricted and each store column can only be reached from the previous or next store columns, except for the first and last store column. The *Worker* can walk to the first and last store columns without any restrictions.

When the *Worker* extends his route from the first or last store column, he ignores all obstacles within his route.

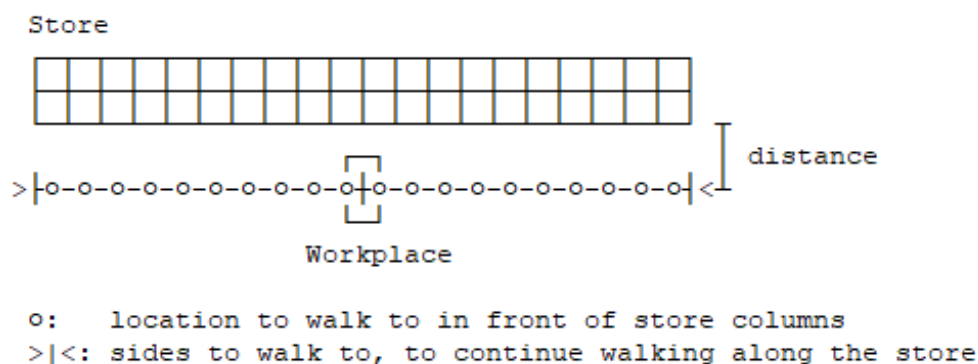
Capacity	-1	
	☒ Worker Stays Here After Completing the Job	
Distance in Queue	1	m
	☐ Walk to the End of the Animation Path of the Queue	
	☐ Walk Along the Animation Path of the Queue	
Pick/Drop at Store	Walk Along Store to Store Column	

For this setting the following features are not available:

- You cannot enter a **Capacity**.
We assume the **Capacity** to be infinite.
- You cannot select **Worker Stays Here After Completing the Job**.
- You cannot enter the **Distance in Queue**.
- You cannot select **Walk to the End of the Animation Path of the Queue**.
- You cannot select **Walk Along the Animation Path of the Queue**.

Plant Simulation determines the locations to which the *Worker* walks by the dimensions of the side of the *Store* at which the *Workplace* is placed, by the number of columns at that side, and by the distance of the center point of the *Workplace* to the *Store* itself.

The diagram of the location to which the *Worker* walks looks like this:



Walk Along Store to Store Column does not increase the number of commonly reachable locations for the route calculation in the same way as the setting **Walk to Store Column** does. Thus, performance impact should be smaller.

SimTalk

PickDropAtStore [SimTalk]

getRouteLength [SimTalk] - Workplace

[getRouteCoordinates \[SimTalk\]](#)

See also

[Model a Worker Picking Parts Up at/Placing Parts in a Store](#)

[Capacity \[text box\] - Workplace](#)

[Worker Stays Here After Completing the Job](#)

[Distance in Queue](#)

[Walk to the End of the Animation Path of the Queue](#)

[Walk Along the Animation Path of the Queue](#)

Tab Times

Tab Times [Workplace]

Define **Times** as described under the **Tab Times**.

The tab **Times** provides these settings:

[Loading Time \[Workplace\]](#)

[Unloading Time \[Workplace\]](#)

[Recovery Time \[Workplace\]](#)

Select a distribution from the drop-down list and type the values that this distribution requires into the text box. *Plant Simulation* shows the parameters along the upper border of the tab. You can also select a constant time (**Const**).

You can set the type of the distribution and a complete set of parameters with the method [setTypeAndAttr \[SimTalk\]](#).

Loading Time [Workplace]

Select a distribution, and type the values which that distribution requires into the text box. *Plant Simulation* shows these values along the upper border of the tab. You can also type in a constant time (**Const**).

Loading Time	Const ▾	2:00	■
Unloading Time	Const ▾	2:00	■
Recovery Time	Const ▾	0:01	■

Remarks

You can set the type of the distribution and a complete set of parameters with the method [setTypeAndAttr \[SimTalk\]](#).

If you select the **Formula distribution**, you can type in a numeric expression or the name of a *Method*. You can use the anonymous identifier @ to access the *part* for which the specified time applies. The anonymous identifier ? points to the *Workplace*. You can access the *Worker* with ?. Cont.

Loading Time

The **Loading Time** is the time it takes the **Worker** for picking up a single *part* at a station. After the **Loading Time** has elapsed, the *Worker* waits for the **Maximum Dwell Time** to pass if he still has carrying capacity. If not, he immediately carries the *part* to the target station and places it there. If the *Worker* is to load more than one *part*, the **Maximum Dwell Time** should at least be as long the carrying capacity times the **Loading Time**.

The *part* is booked on the *Worker* while he picks it up. The processing station is locked until the **Loading Time** has elapsed.

Note

The entrance of the *Workplace* is closed while its **Loading Time** is running as the *Worker* functions with reservations for the *Workplace*.

Behavior During Interruptions of the Worker

Plant Simulation extends the **Loading Time** by **failure times** of the *Worker*. The processing station remains locked during the failure.

If the *Worker* **pauses** during the **Loading Time**, *Plant Simulation* cancels the **Loading Time** and considers it as completely elapsed. This means that the *Worker* carries at least one *part* to the *WorkerPool*.

If the **shift** ends during the **Loading Time**, *Plant Simulation* cancels the **Loading Time** and considers it as completely elapsed. This means that the *Worker* carries at least one *part* to the *parts buffer*.

Behavior During Interruptions of the Processing Station

Failures of the processing station do not affect loading the *part* by the *Worker*. The *part* is already located at the *Worker* and loading takes place during the failure.

The same applies for processing stations which are **stopped**, **paused**, or **unplanned**.

SimTalk

[LoadingTime \[SimTalk\] - Workplace](#)

[LoadingTime \[SimTalk\] - Workplace.Type \[SimTalk\] - distribution parameter](#)

See also

[Formula \[distribution\]](#)

Unloading Time [Workplace]

Select a distribution, and type the values which that distribution requires into the text box. *Plant Simulation* shows these values along the upper border of the tab. You can also type in a constant time (**Const**).

Loading Time	Const ▾	2:00	■
Unloading Time	Const ▾	2:00	■
Recovery Time	Const ▾	0:01	■

Remarks

You can set the type of the distribution and a complete set of parameters with the method [setTypeAndAttr \[SimTalk\]](#).

If you select the **Formula distribution**, you can type in a numeric expression or the name of a *Method*. You can use the anonymous identifier @ to access the *part* for which the specified time applies. The anonymous identifier ? points to the *Workplace*. You can access the *Worker* with ?. Cont.

Unloading Time

The **Unloading Time** is the time it takes the **Worker** for placing the *part* he picked up at another station onto the target station. If he is to unload several *parts*, the **Unloading Time** accrues for each *part*.

The *part* is booked on the *Worker* while he places it on the target station. The target processing station is locked until the **Unloading Time** has elapsed.

Behavior During Interruptions of the Worker

Plant Simulation extends the **Unloading Time** by **failure times** of the *Worker*. The target processing station remains locked during the **failure**.

If the *Worker* **pauses** during the **Unloading Time**, *Plant Simulation* cancels the **unloading time**. The target processing station does not remain locked, and the *Worker* takes the *parts* to the *WorkerPool* with him. After the **pause** is over, unloading starts anew.

If the **shift ends** during the **Unloading Time**, *Plant Simulation* cancels the **Unloading Time** and the *Worker* carries the *part* to the *parts buffer*. The target processing station does not remain locked.

Behavior During Interruptions of the Processing Station Through a Failure, Pause, etc.

The *Worker* only starts unloading when the target processing station can take on the *part* for processing. If this is not the case, *Plant Simulation* enters the *part* to be unloaded into the **Forward Blocking List**. The **Unloading Time** starts after the interruption is over.

The *Worker* continues unloading the *part* when the target processing station is being interrupted during the **Unloading Time**. *Plant Simulation* also enters the part to be unloaded into **Forward Blocking List** if the target processing station cannot accept a *part* at the time of the actual unloading process (after the unloading time has elapsed). The *part* stays with the *Worker*. After the interruption has been removed, the *Worker* deposits the *part* onto the target processing station without any time delay.

SimTalk

[UnloadingTime \[SimTalk\] - Workplace](#)

[UnloadingTime \[SimTalk\] - Workplace.Type \[SimTalk\] - distribution parameter](#)

See also

[Formula \[distribution\]](#)

Recovery Time [Workplace]

Select a distribution, and type in the values which that distribution requires into the text box. *Plant Simulation* shows these values along the upper border of the tab. You can also type in a constant time (**Const**).

Loading Time	Const ▾	2:00	■
Unloading Time	Const ▾	2:00	■
Recovery Time	Const ▾	0:01	■

Remarks

The predefined **Recovery Time** in the class of the *Workplace* prevents that a *Worker* steps onto the *Workplace* while the *Worker*, who is already located on it, has not left it all the way.

Note

You can only define the **Recovery Time** for a *Workplace* that has a **Capacity** of 1. The **Recovery Time** starts once a *Worker* leaves the *Workplace*.

Note

The **Recovery Time** will not take effect, if you set the *Workplace* as the **Home Location** of a *Worker* and if the *Worker* wants to step onto this **Home Location/Workplace**.

You can set the type of the distribution and a complete set of parameters with the method **setTypeAndAttr [SimTalk]**.

If you select the **Formula distribution**, you can type in a numeric expression or the name of a *Method*. You can use the anonymous identifier @ to access the *part* for which the specified time applies. The anonymous identifier ? points to the *Workplace*. You can access the *Worker* with ?. Cont.

SimTalk

[RecoveryTime \[SimTalk\]- Workplace](#)

See also

[Formula \[distribution\]](#)

[Recovery Time \[general description\]](#)

[HomeLocation \[SimTalk\] - Worker](#)

[Capacity \[SimTalk\] - Workplace](#)

Tab Controls

Tab Controls [Workplace]

Click the ellipsis button and select a *Method* in the dialog **Select Object**. Type in the source code of the respective control.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



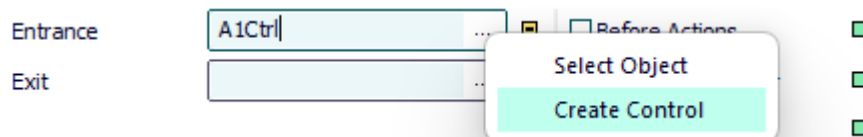
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

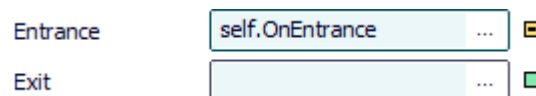
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.

- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

[Entrance Control \[Workplace\]](#)

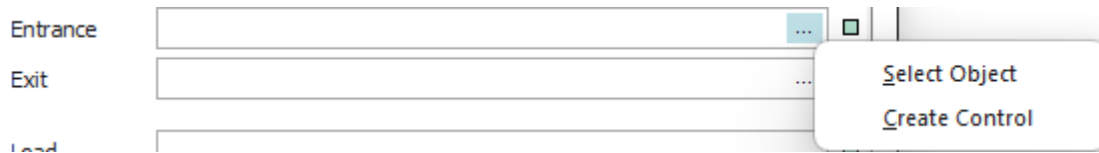
[Exit Control \[Workplace\]](#)

[Load Control](#)

[Unload Control](#)

Entrance Control [Workplace]

Modifies the built-in behavior of the object. *Plant Simulation* calls the **Entrance Control** as soon as the *Worker* has stepped onto the *Workplace*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



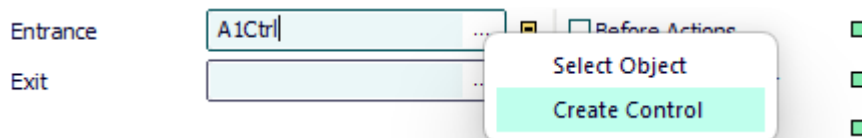
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

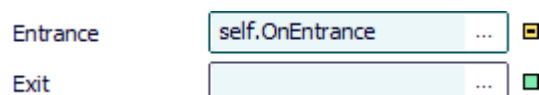
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

[EntranceCtrl \[SimTalk\] - Workplace](#)

See also

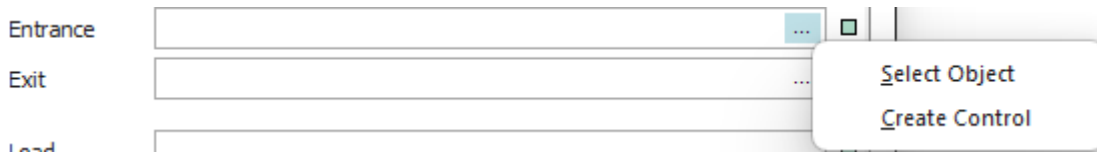
[Select Object \[for controls\]](#)

Video on YouTube

<https://youtu.be/HiPziA8kxc0?si=QuDu7iePtML5x7De&t=908>

Exit Control [Workplace]

Modifies the built-in behavior of the object. *Plant Simulation* calls the **Exit Control** as soon as the *Worker* has stepped off the *Workplace*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



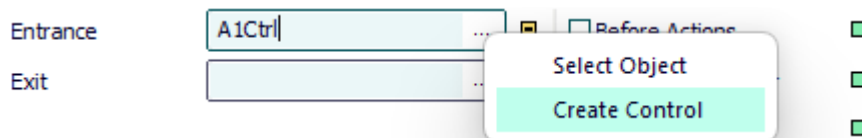
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

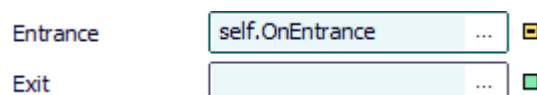
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

[ExitCtrl \[SimTalk\] - Workplace](#)

See also

[Select Object \[for controls\]](#)

Load Control

Modifies the built-in behavior of the object. The object calls the **Load Control** as soon as the *Worker* steps onto the *Workplace*. The **Load Control** sets how the part is transferred from the *work station* to the *Worker* staying on the attached *Workplace*. After picking the part up, the *Worker* carries it to the target object



Remarks

Plant Simulation also calls the **Load Control** when the *Worker* is already located on the *Workplace* and is to load and carry parts to the target object from there.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

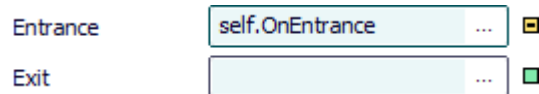
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard load control** as a *user-defined attribute* looks like this:

```
// workplace load control ...
// ... to transfer MUs from the associated station to the worker on the
workplace
// @: worker
// ?: workplace
if @.failed = false and @.pause = false
    // get the first MU to exit and repeat this until no MU is about to exit
    var mu:object := ?.station.cont

    while mu /= void and @.full = false
        // transfer the exiting MU to the worker
        mu.move(@)

        // with configured load time on workplace, wait for EndOfTime event
        causing the transfer to worker
        waituntil mu.~ = @

        mu := ?.station.cont
    end
end
```

SimTalk

[LoadCtrl \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Unload Control

Modifies the built-in behavior of the object. The object calls the **Unload Control** as soon as the *Worker* steps onto the *Workplace* attached to the target station.



Remarks

The **Unload Control** sets how the *Worker* places the part, which he carried to the *Workplace*, onto the attached *work station*.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

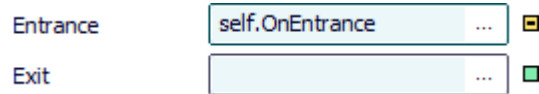
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard unload control** as a *user-defined attribute* looks like this:

```
// workplace unload control ...
// ... to transfer all MUs from a worker on the workplace to the associated
station
// @: worker
// ?: workplace
if @.failed = false and @.Pause = false
    // remember the station associated to the workplace
    var station:object := ?.station

    // get the first MU about to exit to repeat this until no MU wants to
    exit anymore
    var mu:object := @.cont

    while mu /= void and station.full = false
        // wait until the entrance is free (for example for conveyors)
        waituntil station.EntranceFree

        // transfer the MU from the carrying worker to the station
        mu.move(station)

        // with configured unload time on workplace, wait for EndOfTime
        event causing the transfer from worker to station
        waituntil mu.~ = station

        mu := @.cont
    end
end
```

SimTalk

[UnloadCtrl \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

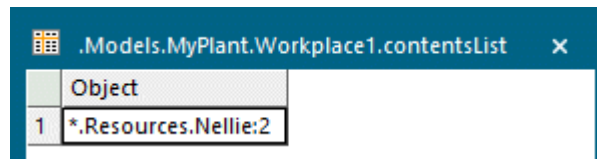
View Menu

The **View Menu** provides commands to access its functions.

Refresh [on View menu]	Contents [Workplace]
Show Statistics Report [on View menu]	Contents and Reserved Places
Show Attributes and Methods [on View menu]	Forward Blocking List

Contents [Workplace]

Opens a list, which shows all *Workers* that stay on the *Workplace*.



The screenshot shows a window titled ".Models.MyPlant.Workplace1.contentsList" with a close button (X). Inside the window is a table with two columns: "Object" and a second column. The first row of the table has the value "1" in the first column and "*.Resources.Nellie:2" in the second column.

	Object	
1	*.Resources.Nellie:2	

Remarks

A *Workplace* with a **Capacity** greater than 1 shows the names and the numbers of the *Workers* that stay on it.

SimTalk

contentsList [SimTalk] - Workplace

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Workplace

_Methods of the Workplace

The *Workplace* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).
- The [Methods of the Material Flow Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				([IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer: =1]) -> object
MUPart	VOID			([Index:integer: =1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).